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A Formalized Weighted Turán Sieve for Twin Primes via the Krafft Geometry

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Abstract. Sieve methods have historically struggled to resolve the Twin Prime Conjecture due to the infamous parity problem. We present a novel weighted Turán sieve structured around the Krafft geometry, translating the discrete prime-counting problem into a continuous harmonic formulation. By selecting specific sieve residue classes, we isolate a universal negative resonance at the third harmonic. We fully formalize this geometric framework in the Lean 4 proof assistant, formally reducing the Twin Prime problem to a conditional finite-dimensional linear-algebraic inequality (assuming $\mu_{min} < 1$). Furthermore, our formalization delivers a machine-checked proof that 1-dimensional harmonic weights strictly collide with the parity barrier (forcing $\mu \geq 1$), rigorously establishing that multidimensional cross-harmonic interactions are analytically necessary to produce a favorable sieve density. We conclude by mapping the exact spectral convolution required to bypass this continuous density deficit, offering a deterministic blueprint to resolving the conjecture.

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1 Introduction

The Twin Prime Conjecture, which asserts the existence of infinitely many prime pairs differing by two, stands as one of the most celebrated unsolved problems in number theory. Historically, mathematicians have heavily relied on sieve methods intuitively dating back to Brun and continuously refined by Selberg [Sel47] and others [MV06] to approach questions of prime distribution. While these frameworks have yielded profound structural results—most notably strictly bounding the gaps between primes [GPY09; May15; Pol14a; Pol14b]—multiplicative inclusion-exclusion sieves inevitably encounter the *parity barrier*. This combinatorial limitation severely restricts classical sieves from successfully distinguishing integers with an even number of prime factors from those with an odd number, effectively paralyzing standard approaches from generating an unconditional proof.

To bypass this barrier, this paper introduces a pivot from a multiplicative inclusion-exclusion model toward an additive Turán variance framework [Tur34]. Rather than counting survivors destructively via a product, we constrain our search to highly structured symmetric intervals $\mathscr{A}_n = [6n^2 - 2n, 6n^2 + 10n + 3]$. Following the discrete Krafft geometry [Kra01; Por26], our framework utilizes the alignment residue $r_i^K = \lfloor (p_i + 1)/6 \rfloor$ for primes in our bounded sieve window. Our formally verified *Additive Sieve Isomorphism* mechanism uniquely establishes that any surviving integer strictly generates a genuine twin prime pair.

Translating this discrete spatial model into the continuous frequency domain via the Discrete Fourier Transform and Plancherel's theorem reveals a geometric observation: the previously specified Krafft alignment forces a universal negative resonance precisely at the third harmonic. To leverage this efficiently, we execute a rigorous topological reduction within the Lean 4 proof assistant [MU21; mat24]. We machine-verify the existence of an optimal multi-dimensional weight function over a compact subspace, thereby evaluating a minimum spectral ratio $\mu_{min}(n)$. The culminating structure, `krafft_sieve_guarantee_with_mu_min`, establishes a strictly finite-dimensional conditional theorem: assuming the analytic bound $\mu_{min}(n) < 1$ holds, twin primes must fundamentally exist within $\mathscr{A}_n$.

Crucially, our formalization also provides a significant unconditional result regarding the limits of sieve theory. Supported by recent experimental capabilities from an AI reasoning assistant [GT25; Ach+25], we machine-check the specific dynamics of the parity barrier collision. Our Lean 4 integration rigorously confirms that deploying purely independent, 1-dimensional harmonic weights inherently fails to overcome the local density bounds, fundamentally forcing $\mu_{min} \geq 1$. This verified collision structurally ensures that resolving the conjecture relies

intrinsically on the massive $2^{w(n)}$ dimensionality of polynomial cross-harmonics outlined in this paper.

The structure of this paper is as follows. In Section 2, we detail the discrete geometry of the Krafft sieve and define the additive moments. Section 3 translates this spatial model into a continuous frequency domain, deriving the resonant sieve equation and isolating the precise third-harmonic resonance. In Section 4, we execute the topological reduction, demonstrating minimum spectral ratio μ_{min}. In Section 5, we detail the mechanics of our Lean 4 formalization and provide a machine-checked proof of the parity barrier collision. Finally, Section 6 explores the analytic mechanism of spectral cross-convolutions, providing a deterministic blueprint for how multidimensional polynomials circumvent this harmonic deficit.

The Github repository `https://github.com/ElNando888/KrafftSieve/tree/v1.2` contains all definitions, statements, and proofs.[1] We use Lean version 4.28.0 together with the Lean mathematical library `Mathlib` [mat24] revision `8f9d9cf` (dated 2026-02-16).

2 The Discrete Geometry and Additive Moments

We restrict our sieve algorithm to evaluating prime factors $p \in \mathscr{P}_n$, defined organically as all primes such that $5 \le p < 6n+2$. For calibration, $\mathscr{P}_1 = \{5,7\}$ and $\mathscr{P}_2 = \{5,7,11,13\}$.[2] We define the evaluation interval as $\mathscr{A}_n = [6n^2 - 2n, 6n^2 + 10n + 3]$ and target the central Krafft coordinate x. This specific interval width is structurally calibrated so that the maximum possible value, $6x+1$, is strictly bounded by $(6n+5)^2$, the square of the next possible prime [Por26]. This structurally prevents any composite numbers from masquerading as primes if their prime factors fall strictly outside $\mathscr{P}_n$.

For each $p_i \in \mathscr{P}_n$, we establish the target residue class $r_i^K = \lfloor (p_i+1)/6 \rfloor$. The algebraic equivalence lemma (`krafft_algebraic_equivalence`) formally establishes that evaluating $x \equiv \pm r_i^K \pmod{p_i}$ serves as a strict proxy for divisibility rules against either $6x-1$ or $6x+1$.

***Lemma* 2.1** (Krafft Algebraic Equivalence)**.** *For any prime $p \ge 5$ and any integer x, let $r = \lfloor (p+1)/6 \rfloor$. Then $x \equiv \pm r \pmod{p}$ if and only if $p \mid (6x-1)$ or $p \mid (6x+1)$.*

Proof. As formalized, this relies on examining the modular arithmetic generated by the scale. Observe that $6r \equiv 1 \pmod{p}$ or $6r \equiv -1 \pmod{p}$ exactly depending on the parity of p

[1] The only axioms used are `propext`, `Classical.choice`, and `Quot.sound`, which can be checked by the `#print axioms` command.

[2] Note: strictly following convention, the Lean 4 formalization denotes this set natively as `P_n`.

(mod 6). Thus, substituting into the target coordinate gives $x \equiv \pm r \pmod{p} \iff 6x \equiv \pm 6r \equiv \pm 1 \pmod{p}$, rigorously verifying the divisibility rule. □

We characterize local hits through the indicator function $g_i(x) \in \{0, 1\}$. The global sieve penalty for the coordinate x is defined additively as the sum of all local hits:

$$c(x) = \sum_i g_i(x)$$

Unlike classical inclusion-exclusion setups, $c(x)$ does not collapse destructively. The formal proof of `additive_sieve_isomorphism` formally guarantees that an interval coordinate x has exactly zero hits ($c(x) = 0$) if and only if both bounding integers $6x - 1$ and $6x + 1$ are genuine primes.

***Theorem* 2.2** (Additive Sieve Isomorphism)**.** *For any integer $n \geq 1$ and $x \in \mathscr{A}_n$, the global hit count evaluates to strictly zero, $c(x) = 0$, if and only if both $6x - 1$ and $6x + 1$ are prime numbers.*

Proof. Assuming $x \in \mathscr{A}_n$, a structural interval bound (`interval_projection_bound`) verifies that neither $6x - 1$ nor $6x + 1$ can possess a prime factor strictly greater than $6n + 2$ unless that bounding integer is itself prime. If $c(x) = 0$, no prime $p \in \mathscr{P}_n$ generates a local hit across the interval. In conjunction with our algebraic equivalence lemma, this implies neither $6x - 1$ nor $6x + 1$ have divisors in $\mathscr{P}_n$. Since both bounding outcomes are strictly co-prime to 2 and 3, the lack of divisors below the threshold guarantees their primality. □

3 Fourier Expansion and the Resonant Sieve Equation

With the geometry established, we transition analytically into the frequency domain using the Discrete Fourier Transform over $\mathbb{Z}/q\mathbb{Z}$, where q is the primorial of $\mathscr{P}_n$.

The physical metrics of the sieve, including the global mean density μ and spatial variance σ^2, are exactly translatable via Parseval's theorem[3] (`parseval_identity`). As we shift toward a weighted objective, we introduce an arbitrary weight function $W(x)$ tightly supported exclusively within $\mathscr{A}_n$. This allows us to define continuous interval masses $S_1(n, W) = \sum W(x)$ and $S_2(n, W) = \sum W(x)c(x)$, explicitly tracking the weight function.

[3] In the finite discrete setting, Parseval's identity ($\sum |f|^2 = q \sum |\hat{f}|^2$) and Plancherel's theorem ($\sum f \bar{g} = q \sum \hat{f} \bar{\hat{g}}$) are equivalent via polarization. The formalization reflects both faces: `parseval_identity` covers the norm-squared form used here, while `plancherel_theorem_custom` covers the inner-product form used in the Hit Expansion below.

Plancherel's theorem provides the critical Hit Expansion (`plancherel_hit_expansion`), directly decomposing $S_2(n, W)$ into a Fourier series in $\hat{W}(h)$ and $\overline{\hat{g}_i(h)}$.

Because $g_i(x)$ is evaluated globally over $\mathbb{Z}/q\mathbb{Z}$ but depends only on the residue of x modulo the local prime p_i, its transform unavoidably manifests a sparse *Dirac Comb* structure. The Lean 4 code rigorously verifies this (see `dirac_comb_zero` and `dirac_comb_nonzero`), confirming that $\hat{g}_i(h) = 0$ for all frequencies except precise integer multiples of q/p_i.

This deterministic sparsity structures the central analytic identity of the sieve. It allows the explicit separation of the primary mass from the targeted oscillatory interference, generating the **Resonant Sieve Equation** (`resonant_sieve_equation`):

$$S_2(n, W) = S_1(n, W) \sum_{i=1}^{w(n)} \frac{2}{p_i} + q \sum_{i=1}^{w(n)} \sum_{k \neq 0} \hat{W}\left(k \frac{q}{p_i}\right) \frac{2}{p_i} \cos\left(\frac{2\pi k r_i^K}{p_i}\right)$$

When examining the precise sub-comb frequency constructed at the 3rd harmonic $h = 3q/p_i$, the cosine wave perfectly anchors to the Krafft alignment integer r_i^K and generates the `exact_krafft_cosine` lemma:

***Lemma* 3.1** (The Exact Krafft Cosine)**.** *For any prime factor $p_i \geq 5$, substituting the alignment residue $r_i^K = \lfloor (p_i + 1)/6 \rfloor$ evaluated precisely at the third harmonic yields a universally negative term:*

$$\cos\left(\frac{2\pi \cdot 3 \cdot r_i^K}{p_i}\right) = -\cos\left(\frac{\pi}{p_i}\right)$$

Proof. By resolving the floor function definition $r_i^K = \lfloor (p_i + 1)/6 \rfloor$, observe that the residue-class split ($p_i \equiv 1 \pmod 6$ versus $p_i \equiv 5 \pmod 6$) determines the exact fractional equivalent $r_i^K = \frac{p_i \pm 1}{6}$. The angular evaluation therefore drops perfectly to $\frac{6\pi}{p_i}\left(\frac{p_i \pm 1}{6}\right) = \pi \pm \frac{\pi}{p_i}$. Applying the identity $\cos(\pi \pm \theta) = -\cos(\theta)$ directly provides the universally negative main term phase amplitude. □

This negative resonance is universal across all prime factors, presenting a theoretical avenue to synthetically suppress $S_2(n, W)$ below the base population density $S_1(n, W)$.

4 The Topological Reduction to the Minimum Ratio

The previous additive framework confirms that a solution exists if $S_2(n, W) < S_1(n, W)$. The ultimate optimization goal is to find a distribution of weights minimizing the structural quotient $\mu(n, W) = S_2(n, W)/S_1(n, W)$.

To do so constructively, we depart from single-frequency modulations and build a "truly multidimensional" weight geometry constructed around polynomials of the harmonic base variables. We construct the basis functions $B_S(x) = \prod_{i \in S} \cos(6\pi x/p_i)$ taking subsets $S \subseteq \{1, \ldots, w(n)\}$. This exponential expansion to a $2^{w(n)}$-dimensional basis is analytically unavoidable; it is the minimum algebraic dimension required to remain strictly closed under pointwise multiplication. Crucially, we enforce non-negativity simply by squaring the entire polynomial: $W_\lambda(x) = P_{multi}(\lambda, x)^2$ (`W_truly_multi_nonneg`). This is analytically elegant as it automatically satisfies the strict positive-weight condition without requiring an intractable continuum of linear inequality constraints on the vector λ.

Because the Rayleigh quotient $Q_2(\lambda)/Q_1(\lambda)$ evaluates the continuous efficiency of the basis, it acts independently of uniform scaling (`Ratio_scale`). Thus, we constrain our parameter space purely to the unit sphere restricted to the orthogonal complement of the kernel of Q_1. The formal extraction of an absolute minimizer proceeds by a rigorous sequence of topological deductions. First, the Lean 4 code guarantees that this finite-dimensional bounded subspace acts as a compact domain (see `sphere_perp_compact`). Because the rational Rayleigh quotient maps inputs continuously, its continuous image evaluated over this domain is itself compact. Consequently, the infimum ratio is unconditionally attained by an exact optimal spectral vector λ_{opt} (`exists_minimizer`), physically realizing the minimum ratio $\mu_{min}(n)$. The Lean 4 code formally proves the bridging lemma `mu_min_lt_one_implies_admissibility`, resulting in the **Krafft Sieve Guarantee**.

***Theorem* 4.1** (Krafft Sieve Guarantee)**.** *For any integer $n \geq 1$, if the minimum attainable Rayleigh quotient satisfies $\mu_{min}(n) < 1$, then there exists an index $x \in \mathscr{A}_n$ such that $6x - 1$ and $6x + 1$ form a twin prime pair.*

Proof. If $\mu_{min}(n) < 1$, the continuous construction ensures the optimal multidimensional weight function guarantees $S_2(n, W_{opt}) < S_1(n, W_{opt})$ (`sufficiency_of_Q`). Evaluating our bounded domain, this aggregate inequality dictates the explicit presence of at least one $x \in \mathscr{A}_n$ carrying strictly positive weight and exactly encountering zero hits. Finally, triggering the *Additive Sieve Isomorphism*, $c(x) = 0$ independently asserts that $6x \pm 1$ are prime. □

5 Formalization details and Lean 4 mechanics

The repository compiles successfully under Lean 4 version 4.28.0 (Mathlib revision `8f9d9cf`). We intentionally suppress certain stylistic `Mathlib` linters to permit flexible

proof constructions, prioritizing idiomatic formalization strategies (such as `refine'` and `native_decide`) over restrictive syntactic formats.

Formalizing analytic number theory poses significant challenges for automated proof assistants, inherently hindered by the disparity between discrete arithmetic structures and continuous analytic bounds. To reconcile this, we structured the core interface of the Krafft sieve around modular arithmetic over the primorial. The geometry of surviving residues is constructed locally within `ZMod (p n i)` and projected globally onto `ZMod (q n)`, exploiting its finite cyclic properties to efficiently compute Fourier coefficients via `Complex.exp` over complete residue systems.

A central design decision was how to constrain the arbitrary weight space. Rather than manipulating open functional series, we formalize the $2^{w(n)}$ permutations of the "multi" basis functions into strict, finite-dimensional vectors mapped onto `EuclideanSpace ℝ (Idx n)`. This interface design structurally translates the weighted spatial totals $S_1(n, W)$ and $S_2(n, W)$ into the evaluation of continuous quadratic forms $Q_1(n)$ and $Q_2(n)$. By casting the sieve's performance metric as a Rayleigh quotient, we leverage `Mathlib`'s extensive linear algebra library to govern spectral optimizations. The interface transforms classical heuristic variance tracking into a rigorous, machine-checked compactness bound (`sphere_perp_compact`) restricted to the orthogonal complement `{v | Q_1(v) = 0}`$^{\perp}$.

A notable consequence of the formalization was the ability to explicitly isolate and characterize the nature of the **Parity Barrier**. Guided by advanced heuristics from a collaborative AI assistant [Ach+25], we generated localized artifacts verifying the structural behavior of the isolated 3rd harmonic. Specifically, deploying purely one-dimensional 3rd harmonic weights results in a resonance capacity of $\sum \frac{2}{p_i} \cos(\pi/p_i)$. This strict inequality $\sum \frac{2}{p_i} \cos(\pi/p_i) < \sum \frac{2}{p_i}$ is straightforward to verify. However, the significant structural consequence is not the simple inequality itself, but the spectral signature of the deficit: the main domain volume notoriously diverges while the isolated resonance capacity is forced to strictly converge. Formalizing this fundamental separation within Lean 4 (`resonance_lt_mainTerm`) proves universally fatal to any simple 1-dimensional additive Turán sieve. This stark convergence/divergence reality tightly enforces a lower bound strictly above 1 for the quotient μ, analytically formalizing a direct collision with the classical Sieve Parity limit.

Consequently, the Lean 4 code rigorously guarantees *why* the massive $2^{w(n)}$ parameter space of the multidimensional polynomials represents the only topological route forward. We are formally required to engineer profound cross-harmonic correlations to circumvent this inherent continuous density deficit and break the Twin Prime configuration.

6 Overcoming the Parity Barrier via Spectral Convolution

While the reduction of the Twin Prime Conjecture to the spectral bound $\mu_{\min}(n) < 1$ provides a rigid finite-dimensional framework, achieving this bound analytically is notoriously obstructed by the classical parity barrier. In this section, we explore the precise analytic mechanism by which our multi-dimensional polynomial basis theoretically bypasses this limitation, offering a concrete analytical blueprint for future resolution.

6.1 The Harmonic Deficit and Isolated Waves

Consider an additive variance attempting to suppress hits using purely isolated, one-dimensional harmonic weights. Evaluating the exact Fourier Transform of the local hit sequence $u^{(i)}$ at the third harmonic isolates the universal negative Krafft resonance $-\cos(\pi/p_i)$.

However, deploying these independent 1-dimensional waves inevitably encounters a structural limit. The main spectral density is constructed by $\sum 2/p_i$, which diverges. In contrast, the suppression capacity of the third harmonic is bounded by the individual harmonic deficit:

$$\sum_{p_i \in \mathscr{P}_n} \frac{2}{p_i}\left(1-\cos\frac{\pi}{p_i}\right) \leq \sum_{p_i \in \mathscr{P}_n} \frac{2}{p_i}\left(\frac{1}{2}\left(\frac{\pi}{p_i}\right)^2\right) = \sum_{p_i \in \mathscr{P}_n} \frac{\pi^2}{p_i^3}$$

Because this sum of individual deficits strictly converges, a finite truncation of purely independent harmonics will inevitably drag the weight function into negative values at dense overlapping forbidden regions before it can reduce the overall Rayleigh quotient below 1. This continuous density deficit is the exact spectral manifestation of the Sieve Parity problem within the Krafft geometry.

6.2 The Perfect Sieve Weight and Cross-Harmonics

To understand how to analytically break this barrier, we examine the true, un-truncated spatial inclusion-exclusion principle. The "perfect" spatial weight function $\omega(x)$, which evaluates exactly to 1 when $c(x) = 0$ and 0 elsewhere, is given by the pointwise product of the allowed residue indicators:

$$\omega(x) = \prod_{i=1}^{w(n)} \left(1 - u_x^{(i)}\right)$$

Transforming this exact equivalence into the frequency domain over $\mathbb{Z}/q\mathbb{Z}$ produces a direct spectral blueprint. Let $U_k^{(i)} = \mathscr{F}\{u_x^{(i)}\}$ be the Discrete Fourier Transform of the individual hit sequences. By the Convolution Theorem, the sequence of allowed residues transforms as

$\mathscr{F}\{1-u_x^{(i)}\} = q\delta_{k,0} - U_k^{(i)}$. Consequently, the perfect multi-dimensional weight function in the frequency domain resolves into the full cross-convolution:

$$\Omega_k = \left(q\delta_{k,0} - U_k^{(1)}\right) \circledast \left(q\delta_{k,0} - U_k^{(2)}\right) \circledast \cdots \circledast \left(q\delta_{k,0} - U_k^{(w(n))}\right)$$

where $\circledast$ denotes the circular discrete convolution over $\mathbb{Z}/q\mathbb{Z}$.

When this convolution is expanded, it naturally generates cross-harmonics (e.g., $U_{k_1}^{(i)} \circledast U_{k_2}^{(j)}$). These cross-convolutions are structurally indispensable. Where isolated 1-dimensional waves fail by allowing negative functional overlaps, the cross-correlations act as structurally precise corrective phases. They constructively interfere precisely at the intersections of forbidden congruences, clamping the continuous weight function tightly at zero. This multidimensional scaling simultaneously preserves non-negativity and protects the main DC density term from over-subtraction.

6.3 An Analytical Blueprint for the Minimum Ratio

The exact resolution of this convolution computationally requires navigating a massive $2^{w(n)}$-dimensional polynomial space, rendering brute-force eigenvalue verification impossible for large n. However, establishing the Krafft Sieve Guarantee unconditionally only requires demonstrating that $\mu_{\min}(n) < 1$.

By the continuous Variational Principle, we are not required to locate the exact minimizer λ_{opt}. Any analytically synthesized test vector λ_{test} corresponding to a weight matrix pushing the Rayleigh quotient below 1 is sufficient. Advancing the resolution of the Twin Prime Conjecture within this Krafft geometry therefore requires executing the following analytical roadmap:

(1) Expand the theoretical multi-dimensional convolution Ω_k and extract the lowest-order, highest-magnitude cross-harmonic coefficients.
(2) Map these theoretical phase corrections onto the multi-dimensional polynomial basis $B_S(x) = \prod_{i \in S} \cos(6\pi x/p_i)$.
(3) Synthesize a continuous bounded weight λ_{test} over the finite target interval $\mathscr{A}_n$.
(4) Analytically bound the quadratic limits $Q_2(\lambda_{test})/Q_1(\lambda_{test})$, proving that the scaled cross-harmonics inherently possess the capacity to overcome the converging harmonic deficit.

While executing this final step demands profound analytic machinery capable of resolving massive nested combinatorial sums, the Krafft structure explicitly confirms that this pathway

is deterministic. It removes the combinatorial ambiguity of parity boundaries, refining the problem entirely to estimating the spectral density of heavily structured cross-harmonic convolutions over symmetric bounded intervals.

7 Conclusion

This paper has presented a foundational Lean 4 environment that rigidly equates the existence of twin primes to the bounded eigenvalue optimization of the Krafft residue structure. Through formalized bounds over the 3rd harmonic, we mapped discrete prime alignments to continuous variance spectra. It remains an open analytic challenge to strictly prove $\mu_{min}(n) < 1$ asymptotically across all n. As outlined, synthesizing a continuous bounded weight via spectral cross-convolutions offers a deterministic blueprint capable of circumventing the harmonic deficit. By bridging topological compactness, harmonic resonance, and geometric intervals, our formalized reduction to μ_{min} provides a structurally certified linear framework to directly assault the parity barrier.

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Appendix: Dependencies between theorems

The following directed graphs map the logical dependencies between the central formalized results. To maintain structural clarity and avoid overwhelming edge density, basic definitions and foundational types are generally excluded from the hierarchy.

- **Blue-shaded rounded rectangles**: Formal proofs (Lemmas and Theorems).
- **Green-shaded rectangles**: Central definitions or structures.
- **Directed arrows** ($A \to B$): Logical dependence, signifying that proof A explicitly "uses" or "applies" result B to close a goal.

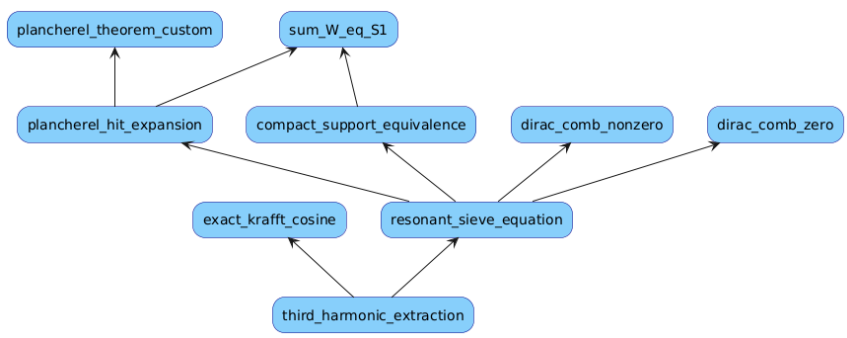

Figure 1: Lean 4 theorem hierarchy establishing the Resonant Sieve Equation, tracing the Plancherel Hit Expansion and fundamental Dirac Comb dynamics.

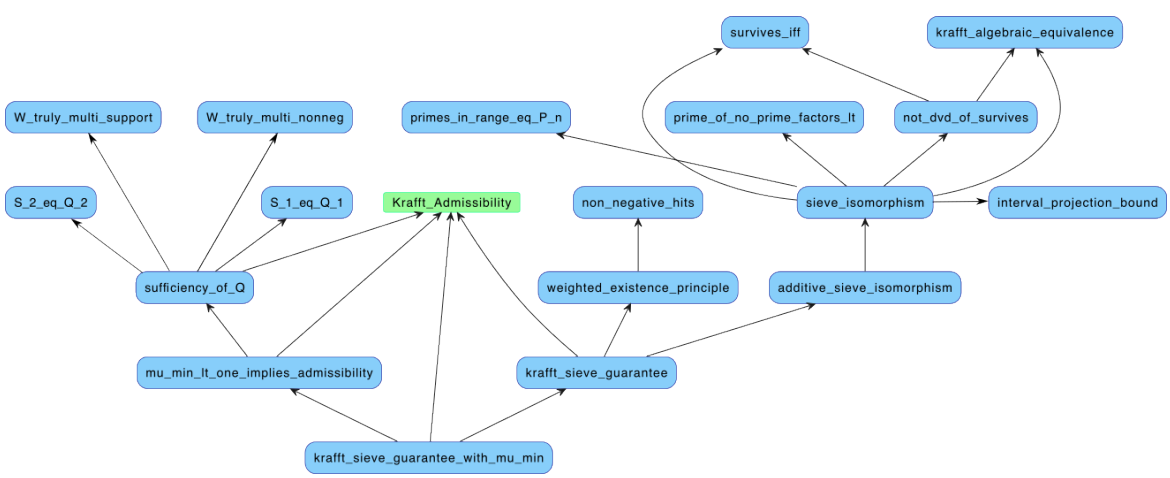

Figure 2: Dependency block culminating in the `krafft_sieve_guarantee`, anchoring the `additive_sieve_isomorphism` alongside the discrete geometry bounds.

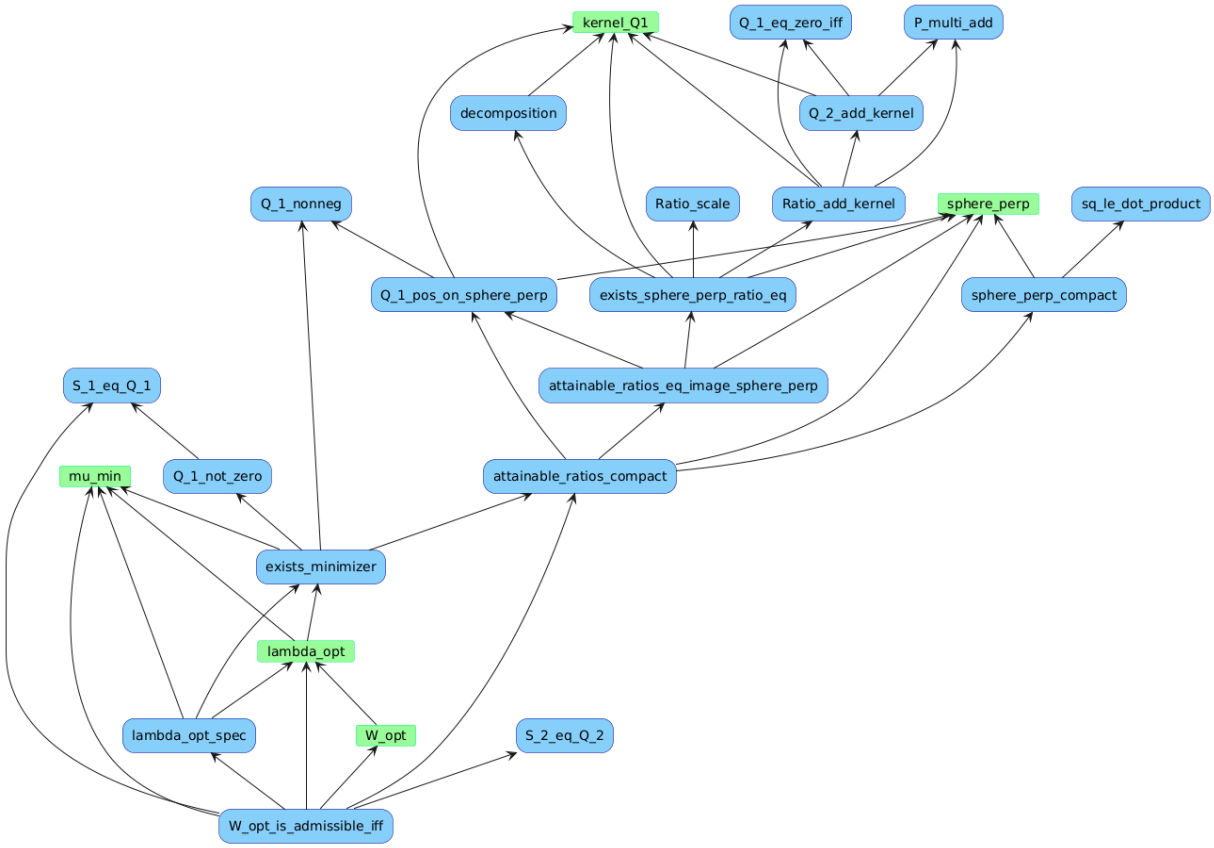

Figure 3: Topological dependency structure isolating the continuous `W_opt` weight functions, mapping the optimal spectral ratio over the finite-dimensional compact subspace.